

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:		Reg. No.:	
Date:		Duration:	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

CO4_AT1

**AI-Based Smart Renewable Energy Asset Performance
Monitoring System**

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DEPT: **BACHELOR OF TECHNOLOGY IN AIDS**

COURSE: **CSA1017 SOFTWARE ENGINEERING**

AI-Based Smart Renewable Energy Asset Performance Monitoring System

1. Introduction

The **AI-Based Smart Renewable Energy Asset Performance Monitoring System** is designed to continuously monitor renewable energy assets such as solar panels, wind turbines, and battery systems. The system collects real-time operational and environmental data through IoT sensors, analyzes the data using AI/ML techniques, predicts possible equipment failures, optimizes maintenance schedules, and generates performance reports.

The purpose of this test case design is to verify the **correctness, functionality, reliability, performance, security, and usability** of the proposed system under normal, boundary, invalid, and exceptional conditions.

2. Functional Requirements to be Tested

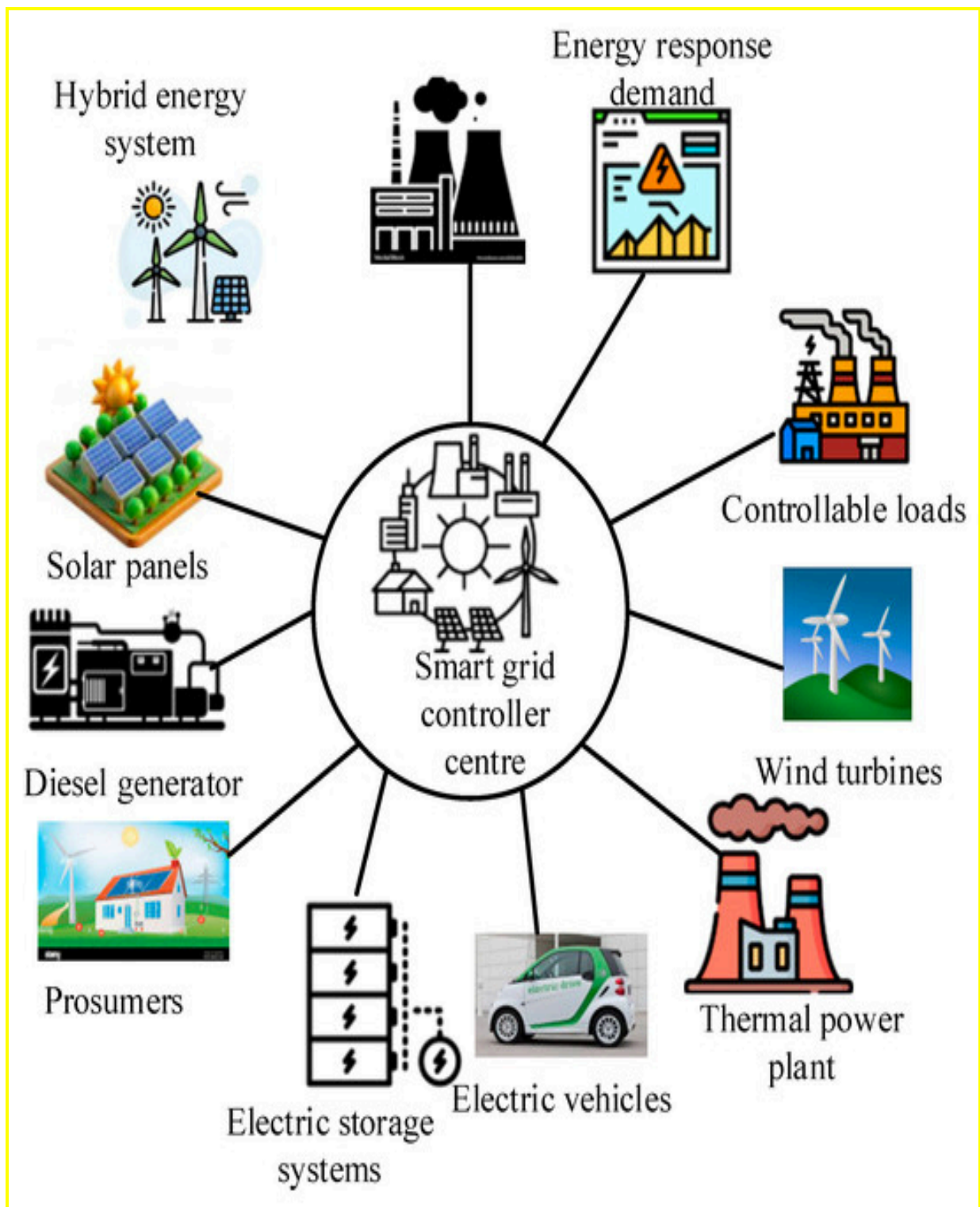
Requirement ID	Functional Requirement
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FR-01	User should be able to log in securely.
FR-02	System should register and manage renewable energy assets.
FR-03	System should receive real-time IoT sensor data.
FR-04	System should monitor asset performance continuously.
FR-05	System should detect abnormal asset behavior.
FR-06	AI model should predict possible equipment failures.
FR-07	System should generate maintenance recommendations.
FR-08	System should analyze weather conditions and their impact on generation.
FR-09	System should display energy-generation information through dashboards.

FR-10	System should generate performance reports.
FR-11	System should provide alerts for critical conditions.
FR-12	System should maintain historical performance data.
FR-13	System should support solar, wind, and battery assets.
FR-14	System should identify underperforming assets.
FR-15	System should allow authorized users to manage system information.

3. Non-Functional Requirements to be Tested

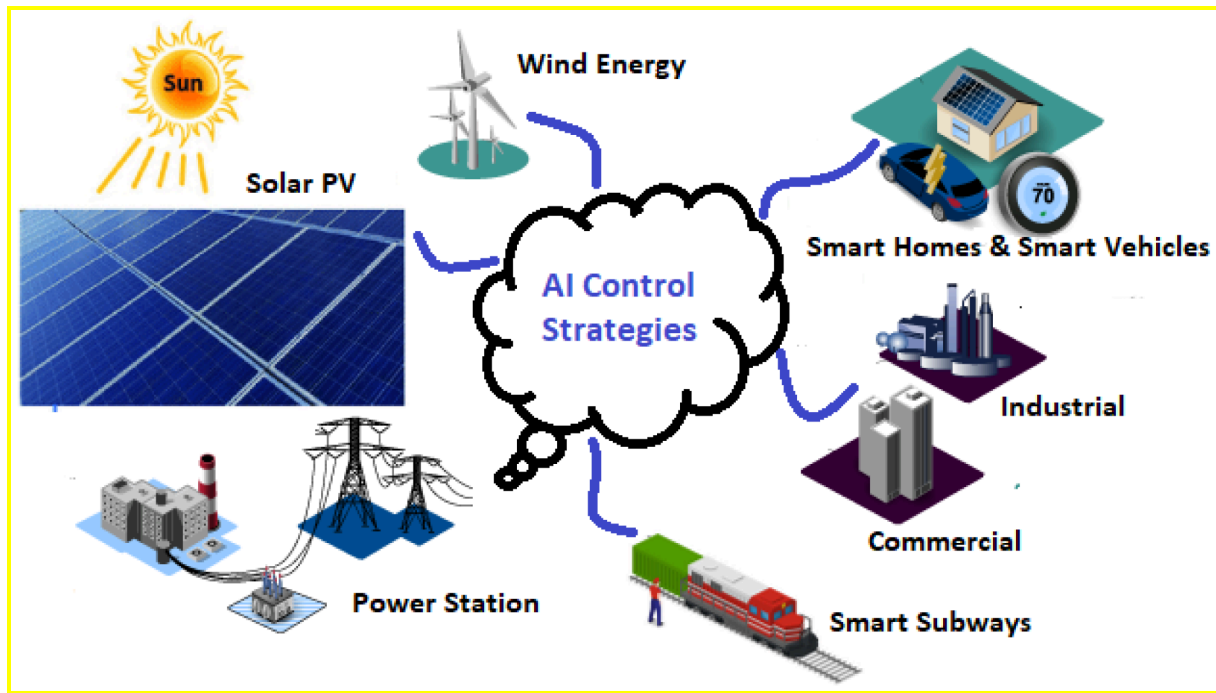
Requirement ID	Non-Functional Requirement
NFR-01	System should provide acceptable response time.
NFR-02	System should support multiple assets and users.
NFR-03	System should remain available during continuous monitoring.
NFR-04	System should protect user and asset data.
NFR-05	System should provide accurate monitoring results.
NFR-06	System should be easy to understand and use.
NFR-07	System should recover correctly from sensor/network failures.
NFR-08	System should maintain data integrity.
NFR-09	System should be scalable as the number of assets increases.
NFR-10	System should provide reliable alerts and predictions.



4. Test Scenarios

Scenario ID	Test Scenario
TS-01	Verify user login and authentication
TS-02	Verify renewable asset registration
TS-03	Verify real-time sensor data collection
TS-04	Verify solar panel monitoring
TS-05	Verify wind turbine monitoring
TS-06	Verify battery system monitoring
TS-07	Verify abnormal-performance detection
TS-08	Verify AI-based failure prediction
TS-09	Verify maintenance recommendations
TS-10	Verify weather-impact analysis
TS-11	Verify performance dashboard
TS-12	Verify alert generation
TS-13	Verify historical data
TS-14	Verify report generation
TS-15	Verify invalid sensor data handling
TS-16	Verify network/sensor failure handling
TS-17	Verify system performance

TS-18	Verify security and authorization
TS-19	Verify usability
TS-20	Verify scalability



5. Detailed Test Cases

5.1 Authentication and User Management

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-01	Valid login	1. Open login page. 2. Enter username and	Valid username/pass word	User successfully logs into dashboard.	To be recorded	Not Executed

		password. 3. Click Login.					
TC-02	Invalid password	Enter valid username and incorrect password. Click Login.	Password: incorrect123		Error message is displayed and login is rejected.	To be recorded	Not Executed
TC-03	Empty login fields	Leave username and password empty. Click Login.	Blank fields		Validation message is displayed.	To be recorded	Not Executed
TC-04	Unauthorized access	Attempt to access admin functions using a normal user account.	Normal user account		Access is denied.	To be recorded	Not Executed
TC-05	Multiple failed logins	Enter incorrect password repeatedly .	5 incorrect attempts		System should apply configured security protection such as temporary	To be recorded	Not Executed

account
logout.

5.2 Asset Registration

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-06	Register solar asset	Enter valid solar asset information and save.	Solar-001, 500 kW	Solar asset is successfully registered.	To be recorded	Not Executed
TC-07	Register wind turbine	Enter valid turbine information and save.	WT-001, 2 MW	Wind turbine is registered successfully.	To be recorded	Not Executed
TC-08	Register battery	Enter valid battery information and save.	BAT-001, 1 MWh	Battery asset is registered successfully.	To be recorded	Not Executed
TC-09	Duplicate asset ID	Register an asset using an existing ID.	Solar-001	System rejects duplicate asset ID.	To be recorded	Not Executed
TC-10	Missing asset information	Leave mandatory fields empty and save.	Missing Asset ID	System displays appropriate validation message.	To be recorded	Not Executed

5.3 Real-Time Monitoring

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-11	Receive valid sensor data	Send valid IoT readings to system.	Temperature=30°C, Power=450 kW	Data is accepted and displayed correctly.	To be recorded	Not Executed
TC-12	Continuous data update	Send sensor readings at configured intervals.	Readings every 1 minute	Dashboard updates with latest readings.	To be recorded	Not Executed
TC-13	Missing sensor value	Send data without temperature value.	Power=450 kW, Temperature=null	System handles missing value without crashing and flags incomplete data.	To be recorded	Not Executed
TC-14	Invalid sensor value	Send non-numeric power value.	Power="ABC"	Invalid data is rejected or flagged.	To be recorded	Not Executed

TC-15	Extreme temperature	Send temperature beyond configured operating range.	Temperature=100°C	System identifies abnormal reading and generates warning/alert according to configuration.	To be recorded	Not Executed
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5.4 Solar Asset Monitoring

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-16	Normal solar performance	Send normal solar readings.	Irradiance=800 W/m ² , Power=450 kW	System displays normal operating condition.	To be recorded	Not Executed
TC-17	Low solar output	Send high irradiance but unusually low power output.	Irradiance=900 W/m ² , Power=100 kW	System detects possible underperformance.	To be recorded	Not Executed
TC-18	Night-time condition	Send zero/near-zero irradiance	Irradiance=0, Power=0	System does not incorrectly report a failure.	To be recorded	Not Executed

e during
night.

TC-19	Panel degradation	Simulate gradual reduction in output.	Output decreasing over time	System identifies performance degradation when configured threshold/model conditions are met.	To be recorded	Not Executed
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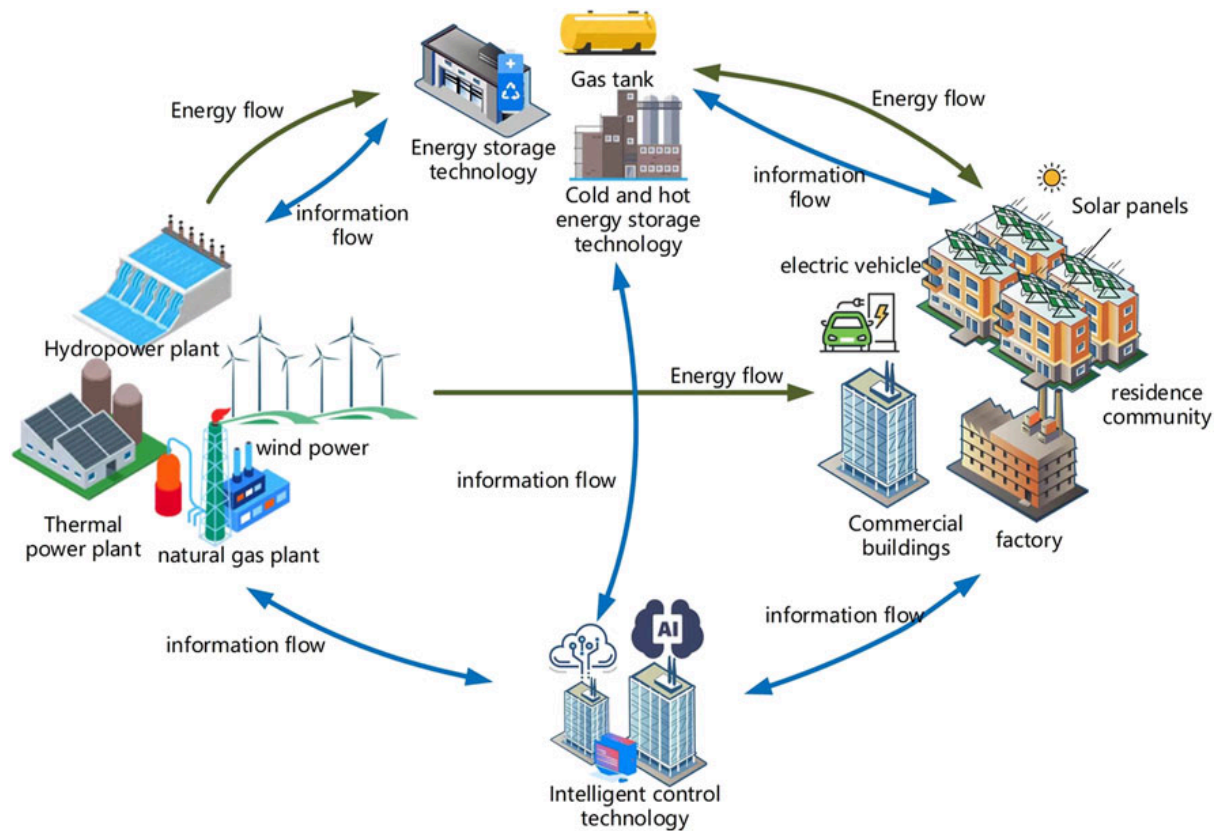
5.5 Wind Turbine Monitoring

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-20	Normal turbine operation	Send valid turbine readings.	Wind=10 m/s, Power=1.8 MW	Normal turbine status is displayed.	To be recorded	Not Executed
TC-21	Low wind condition	Send low wind speed and low power.	Wind=2 m/s, Power=0.1 MW	System does not incorrectly classify the condition as equipment failure.	To be recorded	Not Executed
TC-22	High vibration	Send vibration above configured threshold.	Vibration=High	System detects abnormal turbine condition.	To be recorded	Not Executed

TC-23	Turbine overheating	Send high temperature reading.	Temperature above configured limit	Warning/critical alert is generated according to configuration.	To be recorded	Not Executed
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5.6 Battery Monitoring

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-24	Normal battery condition	Send normal battery readings.	SOC=70%, Temperature=25°C	Battery is shown as healthy/normal.	To be recorded	Not Executed
TC-25	Low battery level	Send SOC below configured threshold.	SOC=10%	System generates appropriate warning.	To be recorded	Not Executed
TC-26	High battery temperature	Send temperature above safe configured threshold.	Temperature=55°C	System generates critical warning/alert according to configuration.	To be recorded	Not Executed
TC-27	Invalid SOC	Send SOC outside valid range.	SOC=150%	System rejects or flags invalid reading.	To be recorded	Not Executed



6. AI-Based Predictive Maintenance Test Cases

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-28	Predict normal asset condition	Provide normal historical and real-time data to AI model.	Normal temperature, vibration and power	System classifies asset as normal/healthy according to model output.	To be recorded	Not Executed
TC-29	Predict possible failure	Provide data representing abnormal	Increasing vibration + decreasing power	AI model identifies elevated failure risk according to	To be recorded	Not Executed

		equipment behavior.		configured prediction criteria.			
TC-30	Failure-risk alert	Generate high predicted failure probability.	High-risk prediction	System generates appropriate maintenance alert.	To be recorded	Not Executed	
TC-31	Insufficient historical data	Run prediction for newly registered asset with insufficient history.	Limited historical readings	System handles insufficient data gracefully and communicates prediction limitations.	To be recorded	Not Executed	
TC-32	AI prediction response	Submit valid monitoring data for prediction.	Complete sensor dataset	Prediction is generated within the defined system response requirement.	To be recorded	Not Executed	

7. Maintenance Scheduling Test Cases

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status	
TC-33	Generate maintenance	Generate high-risk prediction.	High failure risk	Maintenance recommendation	To be recorded	Not Executed	

		recommendation		on is generated.		
TC-34	Normal asset maintenance	Provide healthy asset data.	Low failure risk	No unnecessary emergency maintenance recommendation is generated.	To be recorded	Not Executed
TC-35	Maintenance priority	Generate multiple asset risks.	Asset A: High, B: Medium, C: Low	Higher-risk assets receive higher maintenance priority.	To be recorded	Not Executed
TC-36	Completed maintenance	Mark recommended maintenance as completed.	Asset A	Maintenance status changes to completed and is recorded in history.	To be recorded	Not Executed

8. Weather Analysis Test Cases

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-37	Normal weather	Provide normal	Temperature=28 °C, Wind=8 m/s	Weather information is displayed	To be recorded	Not Executed

	r analysi s	environme ntal data.		and analyzed.		
TC -38	Extrem e weathe r	Provide extreme weather readings.	High wind/rain/temper ature	System identifies relevant environment al conditions and evaluates their impact according to configured logic/model.	To be record ed	Not Execut ed
TC -39	Weathe r impact on solar output	Compare irradiance and solar power data.	Low irradiance + low power	System identifies weather-rela ted reduction where supported by the analysis.	To be record ed	Not Execut ed
TC -40	Missin g weathe r data	Remove weather informatio n.	Temperature=null	System handles missing data without system failure.	To be record ed	Not Execut ed

9. Dashboard and Reporting Test Cases

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
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TC-41	View dashboard	Login and open monitoring dashboard.	Registered assets	Current asset status and performance information are displayed.	To be recorded	Not Executed
TC-42	View historical data	Select asset and date range.	Last 30 days	Historical performance data is displayed correctly.	To be recorded	Not Executed
TC-43	Generate report	Select report period and click Generate.	Monthly report	Performance report is generated successfully.	To be recorded	Not Executed
TC-44	Empty report period	Select a period without available data.	Future date range	System displays appropriate message instead of generating incorrect data.	To be recorded	Not Executed
TC-45	Export report	Generate report and select export option.	Monthly performance report	Report is exported successfully in the supported format.	To be recorded	Not Executed

10. Alert and Notification Test Cases

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-46	Critical equipment alert	Send critical equipment reading.	Temperature above configured limit	Critical alert is generated.	To be recorded	Not Executed
TC-47	Normal reading	Send normal equipment reading.	Normal temperature and power	No false critical alert is generated.	To be recorded	Not Executed
TC-48	Multiple alerts	Generate abnormal conditions for multiple assets.	Solar + wind abnormalities	Alerts are generated for the appropriate assets.	To be recorded	Not Executed
TC-49	Alert acknowledgment	Open and acknowledge an alert.	Alert ID A001	Alert status changes to acknowledged.	To be recorded	Not Executed

11. Boundary Value Analysis

Boundary Value Analysis is used for numerical sensor values and operating thresholds.

Test Case ID	Parameter	Boundary/Test Data	Expected Result
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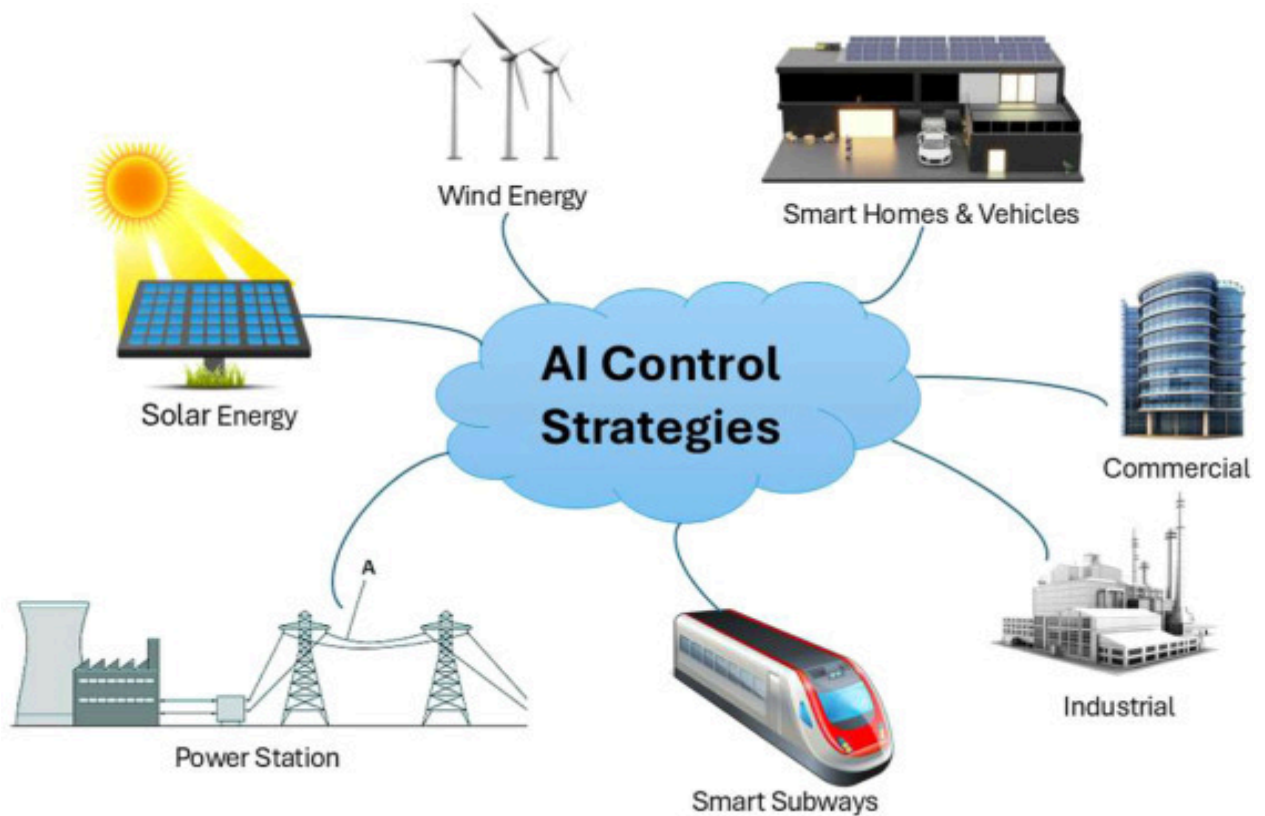
BVA-0 1	Battery SOC	0%	Accepted as valid boundary value if 0% is supported.
BVA-0 2	Battery SOC	100%	Accepted as valid boundary value.
BVA-0 3	Battery SOC	-1%	Rejected/flagged as invalid.
BVA-0 4	Battery SOC	101%	Rejected/flagged as invalid.
BVA-0 5	Temperat ure	Lower operating limit	Correctly accepted and classified according to configuration.
BVA-0 6	Temperat ure	Just below lower limit	Appropriate warning/invalid status according to configuration.
BVA-0 7	Temperat ure	Upper operating limit	Correctly accepted and classified according to configuration.
BVA-0 8	Temperat ure	Just above upper limit	Appropriate warning/critical status according to configuration.
BVA-0 9	Power	0 kW	Correctly handled according to asset operating state.
BVA-1 0	Power	Negative value	Rejected or flagged as invalid where negative generation is not supported.

12. Equivalence Partitioning

The input data can be divided into valid and invalid classes.

Input	Valid Equivalence Class	Invalid Equivalence Class
Asset ID	Unique valid ID	Duplicate/empty ID
Power	Valid non-negative operating value	Negative/non-numeric value

Temperature	Valid sensor range	Impossible/non-numeric value
Battery SOC	0–100%	Below 0% or above 100%
Wind Speed	Valid non-negative value	Negative/non-numeric value
User Login	Correct credentials	Incorrect/empty credentials
Date Range	Valid historical range	Invalid or improperly ordered dates
Sensor Timestamp	Valid timestamp	Missing/invalid timestamp



13. Decision Table Testing

Example: Equipment Alert Decision Table

Condition	Rule 1	Rule 2	Rule 3	Rule 4
Temperature abnormal	No	Yes	No	Yes
Vibration abnormal	No	No	Yes	Yes
Power output low	No	Yes	Yes	Yes
Generate alert	No	Yes	Yes	Yes
Maintenance recommendation	No	Yes	Yes	Yes

Interpretation:

If the equipment shows abnormal temperature, vibration, or low power output, the system should evaluate the condition and generate an alert/maintenance recommendation according to the configured business rules and AI prediction.

14. State Transition Testing

An asset can move through different operational states.

Normal → Warning → Critical → Maintenance → Recovered

Test ID	Case	Current State	Event		Expected State	New
ST-01		Normal	Minor reading	abnormal	Warning	
ST-02		Warning	Condition severe	becomes	Critical	
ST-03		Critical	Maintenance initiated		Maintenance	

ST-04	Maintenance	Fault resolved	Recovered
ST-05	Recovered	Normal monitoring resumes	Normal
ST-06	Normal	Critical failure detected	Critical

The system should not allow invalid state transitions, such as moving directly from **Normal to Recovered** without an appropriate recovery event.

15. Non-Functional Test Cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
NFT-01	Response performance	Load dashboard with monitoring data.	Dashboard loads within the defined performance requirement.	Not Executed
NFT-02	Concurrent users	Simulate multiple authorized users accessing dashboard.	System continues to operate without unacceptable degradation.	Not Executed
NFT-03	Continuous monitoring	Run monitoring continuously for an extended period.	System continues collecting and displaying data without unexpected failure.	Not Executed
NFT-04	Network failure	Disconnect sensor/network connection temporarily.	System handles communication failure gracefully and records appropriate status.	Not Executed

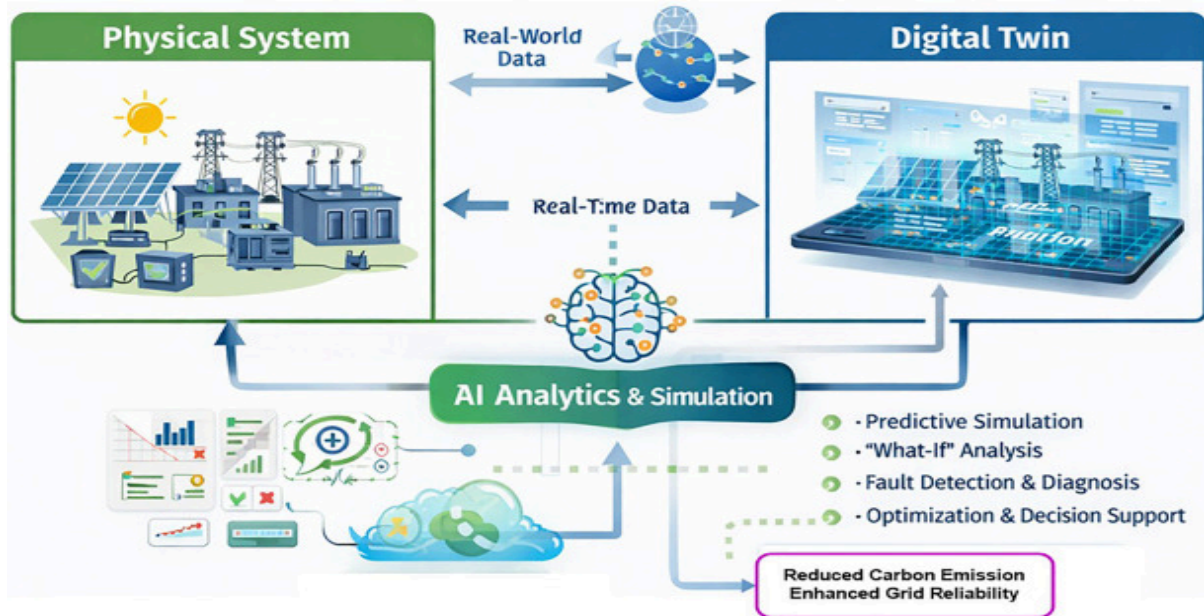
NFT-05	Network recovery	Restore network connection after failure.	System resumes data collection correctly.	Not Executed
NFT-06	Data security	Attempt unauthorized access to asset information.	Unauthorized access is prevented.	Not Executed
NFT-07	Data integrity	Compare received sensor data with stored data.	Data remains accurate and consistent.	Not Executed
NFT-08	Usability	Ask representative users to perform common monitoring tasks.	Users can navigate and perform tasks without significant difficulty.	Not Executed
NFT-09	Scalability	Increase the number of monitored assets.	System continues to support the increased workload according to capacity requirements.	Not Executed
NFT-10	Recovery	Restart monitoring service after controlled failure.	System recovers and resumes monitoring without unacceptable data loss.	Not Executed

16. Requirement Traceability Matrix

Requirement	Covered Test Cases
FR-01 Authentication	TC-01 to TC-05
FR-02 Asset Management	TC-06 to TC-10
FR-03 IoT Data Collection	TC-11 to TC-15

FR-04 Continuous Monitoring	TC-11 to TC-27
FR-05 Abnormal Detection	TC-17, TC-22, TC-23, TC-25, TC-26
FR-06 AI Failure Prediction	TC-28 to TC-32
FR-07 Maintenance Optimization	TC-33 to TC-36
FR-08 Weather Analysis	TC-37 to TC-40
FR-09 Dashboard	TC-41 to TC-42
FR-10 Reports	TC-43 to TC-45
FR-11 Alerts	TC-46 to TC-49
FR-12 Historical Data	TC-42
FR-13 Multiple Asset Types	TC-16 to TC-27
FR-14 Underperformance Detection	TC-17, TC-19
FR-15 Authorization	TC-04
NFR-01 Performance	NFT-01
NFR-02 Scalability	NFT-02, NFT-09
NFR-03 Availability	NFT-03, NFT-10
NFR-04 Security	NFT-06
NFR-05 Accuracy	NFT-07
NFR-06 Usability	NFT-08
NFR-07 Recovery	NFT-04, NFT-05, NFT-10
NFR-08 Data Integrity	NFT-07
NFR-09 Scalability	NFT-09
NFR-10 Reliability	NFT-03, NFT-10

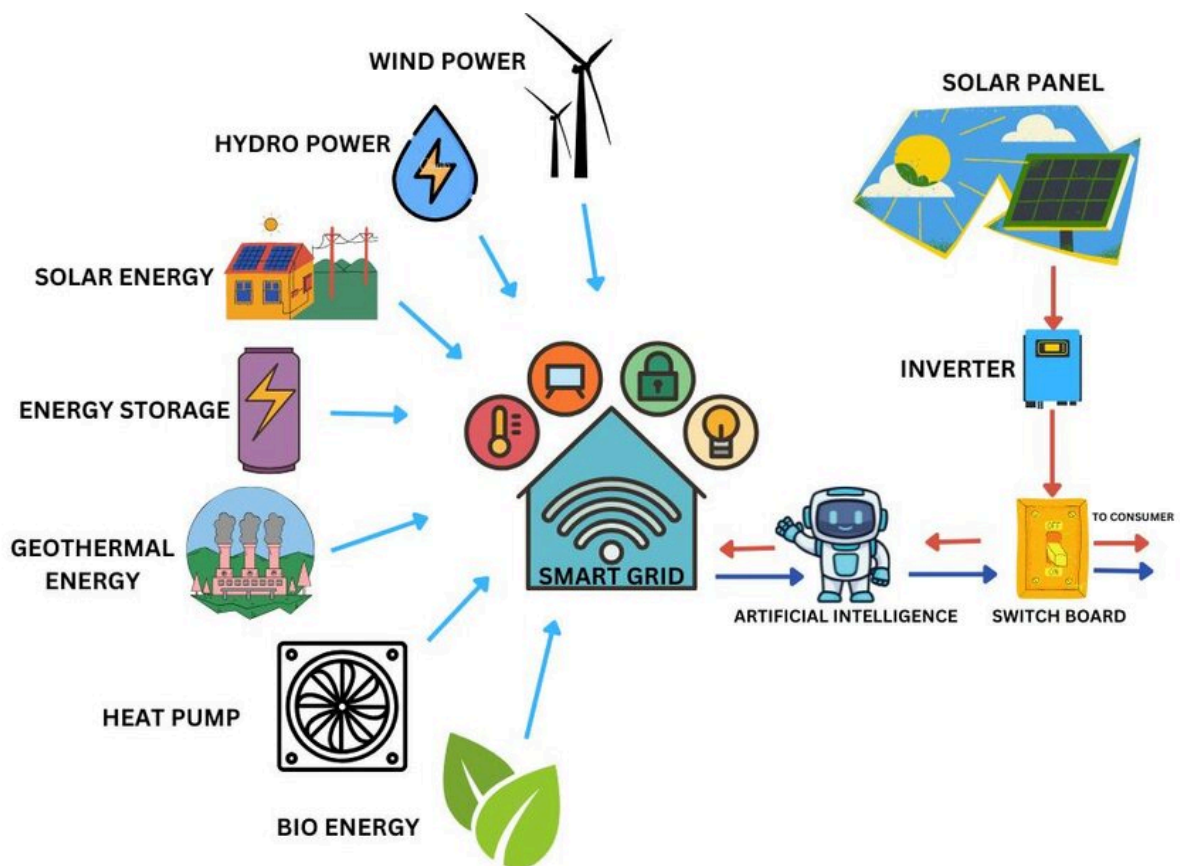
AI and Digital Twin Integration for The Solar Energy and Microgrid



17. Test Coverage Summary

Testing Area	Number of Test Cases
Authentication & Authorization	5
Asset Registration	5
Real-Time Monitoring	5
Solar Monitoring	4
Wind Monitoring	4
Battery Monitoring	4
AI Predictive Maintenance	5
Maintenance Scheduling	4

Weather Analysis	4
Dashboard & Reports	5
Alerts & Notifications	4
Boundary Value Analysis	10
Equivalence Partitioning	8 input classes
State Transition Testing	6
Non-Functional Testing	10
Total Detailed Test Cases	75+



18. Testing Techniques Applied

Equivalence Partitioning

Used to divide sensor and user inputs into valid and invalid categories, reducing the number of test inputs while maintaining good coverage.

Boundary Value Analysis

Used for values such as battery State of Charge, temperature, power, and other operating limits to verify behavior at and around boundaries.

Decision Table Testing

Used to verify combinations of abnormal temperature, vibration, and power conditions and determine whether alerts or maintenance recommendations should be generated.

State Transition Testing

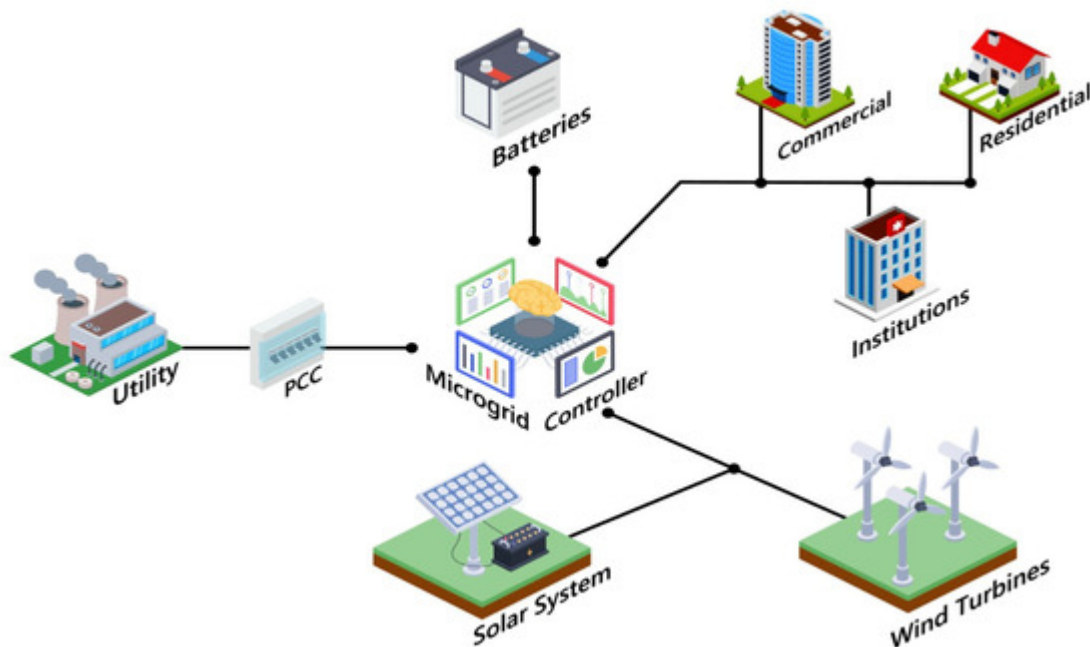
Used to verify asset lifecycle states such as **Normal, Warning, Critical, Maintenance, and Recovered**.

19. Overall Expected Testing Outcome

The testing process should demonstrate that the AI-Based Smart Renewable Energy Asset Performance Monitoring System can:

- Monitor renewable energy assets continuously.
- Collect and validate real-time IoT data.
- Correctly monitor solar, wind, and battery systems.
- Detect abnormal equipment conditions.
- Predict potential failures using AI/ML models.
- Generate appropriate maintenance recommendations.
- Analyze the influence of weather on energy production.
- Generate accurate dashboards and reports.
- Produce reliable alerts for critical conditions.
- Handle invalid, missing, boundary, and exceptional data.

- Maintain security and data integrity.
- Recover appropriately from sensor and network failures.
- Provide acceptable performance, scalability, reliability, and usability.



Conclusion

The proposed test case design provides comprehensive functional and non-functional coverage for the **AI-Based Smart Renewable Energy Asset Performance Monitoring System**. By combining normal, boundary, invalid, and exceptional test conditions with **Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing**, the testing process can identify defects before deployment and ensure that the system reliably supports predictive maintenance, performance optimization, reduced downtime, and efficient renewable energy generation.